

CLOUD-5 — #335: expired and revoked keys stop reading as "Valid"

Batch of 2026-08-26 · merge order 5 of 5 (wire-format change — lands last so the protocol bump sits on top of a quiet main).

```
task_id: gv-335-cloud-5
issue: 335
branch: cloud/335-signature-status
base: main          # rebase onto latest main before opening the PR
adr_number: 0088     # ASSIGNED — wire-contract change; it gets an ADR.
sign_commits_as:
  name: Claude_Max
  email: 262510778+tom2025b@users.noreply.github.com
  # per-commit, ALWAYS: git -c user.name=Claude_Max -c
  user.email=262510778+tom2025b@users.noreply.github.com commit ...
allowed_paths:
  - crates/git-vista-protocol/src/**
  - crates/git-vista-server/src/handlers/tags.rs
  - crates/git-vista/src/**          # ONLY the tag-band display of the new
  statuses; nothing else in the frontend
  - docs/adr/
forbidden_paths:
  - crates/git-vista/src/features/stash/** # another crew's lane today —
  absolutely not
  - ci/browser/**
deliverables:
  - branch pushed, PR opened with "Closes #335"
  - ADR 0088 + index entry
```

Environment truths — read before your first test run

- **~320 server tests fail in your container on unmodified main** (no Landlock). Baseline first; only the diff is yours; both counts in the PR body.
- **cargo build -p git-vista-server --bin gv-sandbox before any server test run.**
- **The browser leg cannot run in your container, and your change touches frontend display** — so this line in the PR body is load-bearing, not boilerplate: "ci/browser/run.sh unrun — cloud container; frontend tag-band display needs the owner's browser-leg run before merge."
- Frontend compile check that DOES work in your container: `cargo clippy -p git-vista --target wasm32-unknown-unknown -- -D warnings` (the wasm target is installed by rustup in the container; if it is not, `rustup target add wasm32-unknown-unknown` first, and say so).

The defect (found by adversarial review of #237, filed as #335)

`classify_verify_tag_output` (`crates/git-vista-server/src/handlers/tags.rs`) maps `gpg` status lines onto `SignatureStatus` — five variants: `Unsigned`, `Valid`, `Invalid`, `UnknownKey`, `Unverifiable`. Two real outcomes have nowhere to go:

gpg status	Meaning	Today reports
<code>EXPKEYSIG</code>	crypto passed; signing key has since expired	<code>Valid</code>
<code>EXPSIG</code>	crypto passed; the signature itself expired	<code>Valid</code>
<code>REVKEYSIG</code>	signature by a REVOKED key — often compromise	<code>Unverifiable</code> (unhandled fallthrough)

An expired key reading identically to a live trusted signature, and a revoked key reading as a shrug, are both honesty defects in a surface whose whole point is stating what the crypto actually said.

Re-verify the classifier and the enum against your checkout before building — variants or wording may have moved since the issue was filed.

The job

1. **Extend `SignatureStatus`** with distinct variants for the three (the issue's framing suggests `ValidExpiredKey` / `ValidExpiredSignature` — or one `Expired { what }` — and `Revoked`; pick the shape that keeps the frontend match exhaustive and the ADR argues it). This is a wire-format change: follow whatever versioning discipline the protocol crate's module docs prescribe (read how #506's v7 bump was done and match it — including whether a bump is required for an additive enum variant, which the protocol docs answer; do not guess).
2. **Classify the three `gpg` statuses** in `classify_verify_tag_output`, killing the `REVKEYSIG` fallthrough explicitly.
3. **Frontend tag band:** display the new statuses with wording that says what happened — "signed, key since expired" is information; "expired" is ambiguity. Revoked reads as a warning, not a shrug. Touch ONLY the tag-band display path.
4. **Tests:** classifier tests from verbatim `gpg` status-line fixtures for all three; an exhaustiveness guard so the NEXT unhandled `gpg` status cannot silently fall through to `Unverifiable` again (if the current code's fallthrough is deliberate, the guard is a test pinning the full list of statuses it may absorb).
5. **Mutation-prove two different ways:** revert `REVKEYSIG` to the fallthrough; then swap the two expired classifications. Red at different assertions, byte-identical restore verified.

Acceptance

1. Three statuses classify distinctly; wire change follows the protocol crate's own bump discipline, ADR 0088 records shape + reasoning.

2. Mutation evidence in the PR body (two red lines, verbatim).
3. `cargo fmt --all` · clippy native AND wasm targets `-D warnings` · server suite zero new failures vs baseline · protocol crate tests green.
4. PR body: baseline counts, the load-bearing browser-leg line, ADR summary, your session tag.

Written by fable · 2026-08-26 · truth-check the classifier against your checkout before building.